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Staging situated negotiation games for (re) designing local healthcare facilities and services

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ABSTRACT

Denmark's ageing population and the accompanying rise in chronic conditions present significant challenges to an already fragmented healthcare system. This study investigates how negotiation games can enable co-exploration, co-analysis, and co-design to support cross-sector collaboration in Danish Community Health Centres (CHCs). Data collected during site visits, interviews, and workshops with healthcare professionals and managers across Denmark demonstrate that negotiation games developed using the staging negotiation spaces (SNS) co-design framework foster a co-analysis process through meaningful dialogue while aligning agendas and building consensus. This iterative approach encourages double-loop learning, where empirical findings from each negotiation space inform subsequent stages. The results highlight the adaptability of various game types—configuration, ranking, and process-oriented—to suit different research and design phases, from initial exploration to collaborative analysis and development. Facilitators play a crucial role in maintaining engagement through challenges, such as struggling with digital formats. This research contributes to health design literature by showcasing how tailored negotiation games based on the SNS framework can bridge sector divides and promote co-created solutions in complex healthcare settings.

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Introduction

Denmark, like many other countries, is facing unprecedented challenges associated with an ageing population. Many individuals are living longer with multiple chronic diseases that require continuous care and treatment.

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Ensuring high-quality care can be difficult in a fragmented healthcare system, as 'these individuals often have multiple care demands and require complex and continuous services spanning professions, sectors and political levels' (Lyngsø, Godtfredsen, and Frølich 2016, 1). In Denmark, responsibilities for care are divided among private general practitioners (GPs), regional hospitals, and municipalities. Danish practitioners have identified 'health policy issues, deficits in legislation, and incentive structures of healthcare' as some of the main system-level factors causing fragmentation (Noor, Gulis, and Eklund Karlsson 2022, 8).

In response, the government provided funding to build local community health centres (CHCs) to foster collaboration between sectors by providing specialist and primary care services at the same location (Sundheds- og Ældreministeriet 2018). In an interview, a local politician explained that she hoped providing services under the same roof would increase efficiency and make it easier for patients to navigate healthcare services. However, to her knowledge, cohesion has not yet happened. CHCs thus have not fulfilled their potential to foster collaboration between Danish regions, municipalities, and healthcare practitioners.

Indeed, cross-sectoral collaboration is a so-called 'wicked' problem (Rittel 1988; Rittel and Webber 1973) that is difficult to address. Often, wicked problems arise when competing institutional logics (Bryson, Crosby, and Stone 2006) counteract economic, regulatory, managerial, and IT infrastructure frameworks. Collaborative design strategies have proven effective in understanding and proposing solutions to wicked problems (Buchanan 1992; Dorst 2011), particularly in healthcare (Nakarada-Kordic et al. 2021). The design tradition offers collaborative methods that embrace and tackle the inherent complexity of healthcare systems by involving diverse user groups, including patients, healthcare professionals, and administrative staff, allowing them to voice their experiences, interests and concerns (Pedersen 2020).

Co-design in healthcare

In a healthcare setting, co-design is a method to design better experiences for patients, their caregivers, and care professionals, as well as new healthcare products, services, and systems that address stakeholders' needs, experiences, and concerns (Sanz, Acha, and García 2021; Shah and Robinson 2008). For instance, Loeliger et al. (2023) co-designed a cancer nutrition care pathway in collaboration with patients, caregivers, and healthcare professionals. Batalden et al. (2016) suggested that such co-production strategies may offer a new perspective on healthcare services. Likewise, Pedersen and Brodersen (2023) demonstrated the importance of staging and facilitating involvement and engagement among users and other central actors in collaborative negotiations to develop healthcare services and systems. Nevertheless, designers and healthcare professionals

sometimes struggle to find a shared language, have different agendas, and therefore have different motivations (Nakarada-Kordic et al. 2021).

Games can help overcome these challenges by fostering dialogue, collaboration, and alignment and reducing the complexity of reality (Brandt 2006; Kim and Nam 2022; Spitters et al. 2017). They are particularly valuable tools in healthcare design, as they engage patients and staff in co-design activities when *staged and facilitated effectively* (Pedersen and Brodersen 2023)

Design games: participatory design

Design games stem from participatory design traditions and are methods or tools based on game logic used in collaborative design settings (Ehn 1988; Pirinen and Tervo 2020; Vaajakallio 2012). In contrast to typical user-centred ethnographic methods of engaging with users (e.g. observations and interviews, which rely solely on visual impressions or verbal communication), design games may be used to foster collaboration, focus dialogue on exploring and investigating current challenges, and point towards future solutions in the form of new products, services and systems (Brandt 2006; Brandt, Messeter, and Binder 2008). For instance, various design games may be used throughout the design process to foster dialogue, collect data during fieldwork (Kjaersgaard, Knutz, and Markussen 2021), and generate ideas with users and other actors central to the investigated problem and solution. It allows participants to voice tacit knowledge and encourages creativity between designers, users, experts, public authorities, and other important actors (Brandt 2006; Ehn 1988; Pirinen and Tervo 2020; Vaajakallio 2012).

Policy games: policy development

Researchers also actively use games in other fields, including public policy and planning. Policy games are a subset of serious games focused on exploring, understanding, or communicating public policy, planning, and governance issues in real-life situations (Kim and Nam 2022; Spitters et al. 2017). Policy games, for instance, encourage reframing existing policies and practices by simulating scenarios and decision-making processes. Stakeholders such as healthcare professionals, patients, policymakers, and other relevant actors can engage in structured dialogue (Duke and Geurts 2004; Mayer 2009). By leveraging the dynamics of play, policy games can help demystify intricate healthcare challenges by, for example, allowing participants to test scenarios (Olejniczak, Wolański, and Widawski 2018) or develop implementation plans (Spitters et al. 2017). Thus, researchers oftentimes use policy games to foster co-analysis of current and future policy scenarios.

In summary, both design games and policy games are effective tools for fostering playful dialogue about serious matters. However, their use settings

and aims are different. Design games, which have their roots in participatory design, are great tools for *collaboratively exploring problems and solutions* iteratively throughout the design process (Brandt, Messeter, and Binder 2008; Dorst and Cross 2001). In contrast, policy games are often played during specific events and are typically used in stand-alone workshop settings where key stakeholders *co-analyse* to *test and implement already identified policy solutions* (Spitters et al. 2017).

Negotiation: a central aspect of both design and policy games

While design games and policy games have different origins and purposes, negotiation is a central aspect of both. Therefore negotiation is at the centre of Figure 1. For instance, scholars from the participatory design tradition noted that 'negotiation seems to be central to all design games, especially in codesign, which builds upon several viewpoints, opinions and skills' (Vaajakallio and Mattelmäki 2014, 66). Likewise, scholars from the policy game tradition reported that 'the main learning experience … was the importance of interaction and negotiation as part of the communication process' (Spitters et al. 2017, 44).

This focus on negotiation is not surprising, since negotiation is a core aspect of collaborative design and innovation (Pedersen et al. 2022). For instance, Buur and Matthews (2008) pointed to negotiation as pivotal for

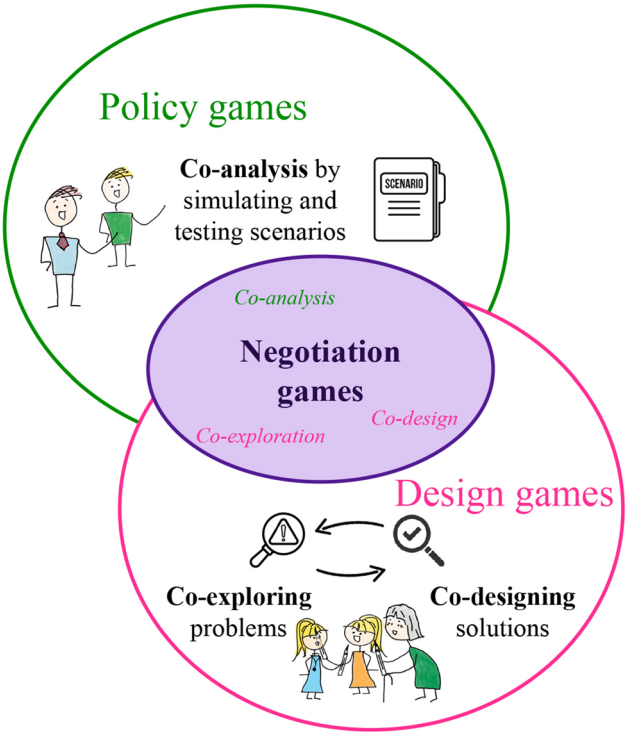


Figure 1. Negotiation is central in both design and policy games.

knowledge transformation and co-creation. Negotiation also plays a central role in the constructivist actor-network theory (ANT) tradition, as it allows us to understand design and innovation as a network alignment process (Law & Lodge, 1984). Negotiation should not necessarily lead to consensus, but foster collaboration and alignment (Star. 2010). A constructivist view that gives equal attention to humans and non-humans enables us to understand negotiation as something that may take place between people, but also between, for instance, a designer and her design (Latour 2008). Inspired by this, Pedersen (2020) combined the theoretical lens of ANT and the actionable aspects of co-design to develop Staging Negotiation Spaces (SNS), a co-design framework which focuses on how to stage and facilitate negotiations in co-design. In a case study centred on developing products to support the swift recovery of stroke patients, Pedersen portrayed the designer or researcher as a humble stage director who sets the stage for negotiation. The designer accomplishes this by creating an environment to explore and identify existing problems while working towards innovative solutions for the future.

The healthcare system, with all its complexity and fragmentation, is a perfect arena for staging negotiations between municipalities, regions, GPs, patients, and others. Thus, when investigating the role of CHCs in terms of their functions, means of collaboration, and role in the healthcare system, it is essential to engage healthcare professionals and managers (from both municipalities and regions) in negotiations about the current state of the system, researchers' preliminary findings, and future solutions.

Here, we conceptualize *negotiation games* as a specific tool aimed at fostering processes of co-exploration, co-analysis, and co-design. Negotiation games can be staged and facilitated to support these aims at different moments during the design process (see Figure 1).

Depending on the specific moment in the design process, a negotiation game may aim to foster one or several of these situated processes. Through a case study investigating the role of CHCs in Denmark, we illustrate how the SNS framework (Pedersen 2020; Pedersen, Bogers, and Clausen 2022) is instrumental in developing and facilitating negotiation games.

This leads us to the following research question:

- How can a staging approach support researchers, designers and health professionals in developing and using negotiation games to foster co-exploration, co-analysis, co-design and alignment across sectors and professions?

Methods

From 2019–2021, we investigated the role of CHCs in local Danish health services as part of a project funded by Helsefonden (Foundation for Health) using an

exploratory approach. During this period, we visited 32 locations across all 5 Danish regions and conducted 38 interviews with 63 informants, including health professionals and managers. Five regional workshops with 3–15 participants were facilitated, and one final conference/workshop was conducted online because of the COVID-19 pandemic. Negotiation games were developed and used as a dialogue tool for exploration, co-analysis, and co-design throughout the project. We used the SNS framework (Pedersen 2020) prescriptively as a strategic dialogue tool to develop, stage, and facilitate games during interviews and workshops. All interviews and workshops were recorded and transcribed.

Staging negotiation spaces

The SNS co-design framework is actionable and analytical. It is based on a theatrical metaphor and it draws upon the participatory design tradition and ANT (Latour 1987; Law 1992). Participatory design provides methods, such as games, for engaging multiple actors in negotiations and introduces staging as a central co-design aspect (Clausen et al. 2020). ANT provides an analytical lens and a vocabulary for understanding the role and performativity of games in such negotiations.

Through the lens of the SNS framework (Figure 2: Illustration of a modified version of the SNS framework, including configuring aspects of the space (1–4 on the right) and reflective questions to ask oneself at each step of the staging process (a–c on the left)), design or research processes are viewed as consisting of a series of negotiation spaces. These discursive spaces are typically configured by (1) objects such as games, (2) a facilitator, (3) the material arrangement of the room or setting (either online or offline), and (4) relevant actors. Pedersen, Bogers, and Clausen (2022) described a series of staging moves that allow designers, researchers and health professionals to reflect upon how to effectively develop and facilitate negotiation games that align with the other configuring elements. These staging moves entail:

a. *Interpreting* the problem/situation (e.g. do current CHCs succeed in creating cohesive care pathways) and *(re)framing* negotiations to motivate specific discussions (e.g. about the role of CHCs in the Danish healthcare system);
b. Producing objects by *inscribing* interpretation and framing into different 'props' (in our case, negotiation games intended to facilitate mutual learning and dialogue); and
c. *Inviting* relevant actors to the negotiations, such as managers, doctors, physiotherapists, and nurses, and *facilitating* their participation in the negotiation games.

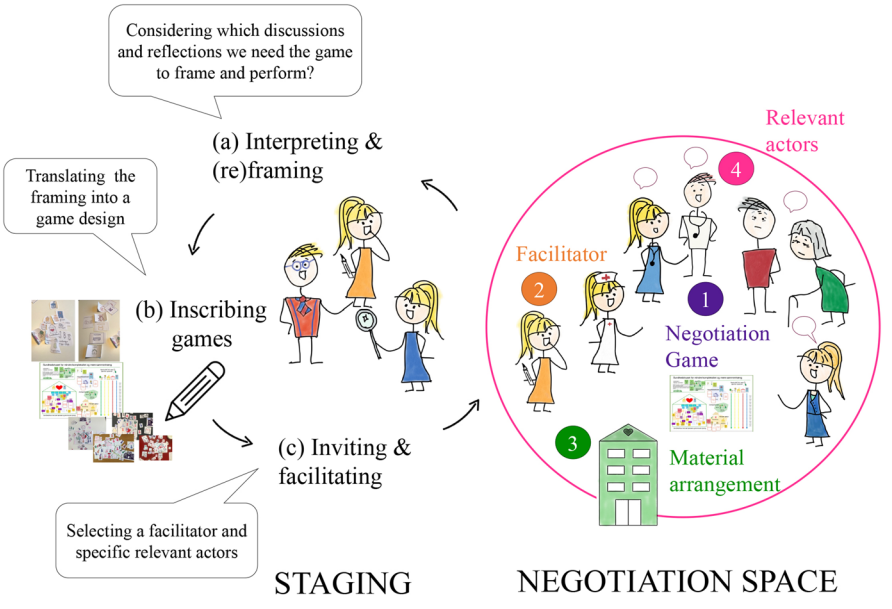


Figure 2. Illustration of a modified version of the SNS framework, including configuring aspects of the space (1–4 on the right) and reflective questions to ask oneself at each step of the staging process (a–c on the left).

During our research project, we investigated how co-creation can be leveraged to support cross-sectoral collaboration in Denmark's fragmented healthcare system. Employing the staging moves in the SNS framework, we developed negotiation games to facilitate negotiations between central actors from the CHCs. We used the games strategically to facilitate these negotiations and allow for mutual learning among healthcare staff from regions and municipalities, general practitioners (GPs), designers, and managers. In this article, we illustrate how a collaborative staging approach (i.e. SNS) can be leveraged to design and facilitate multiple situated negotiation games to support co-exploration, co-analysis, and co-design.

Three types of negotiation games

Throughout the project, we developed three types of negotiation games: configuration games, prioritization games, and process-oriented games (see Figure 3).

While configuration, prioritization, and process-oriented games may take various forms and use different rules, all three types share specific game elements that help structure negotiations according to the project's methodological purpose (Table 1). Game boards, pieces, and question cards serve as structuring elements that create a shared physical or digital space. This arrangement fosters focused discussions and collaborative reflections among participants. These elements' visual and tangible aspects enable participants to visualize and co-construct insights in real time.



Figure 3. Three types of negotiation games.

Table 1. Types of negotiation games developed in the project.

Negotiation Game Type	Purpose	Game-Like Elements Invoked
Configuration games	Co-exploration, Co-analysis, Co-design	Goals (mapping, configuring systems), roles (stakeholders as participants), rules, game board and pieces, visual feedback, exploratory facilitation, scenario-based narratives, iteration, card-drawing
Prioritization games	Co-analysis	Goals (ranking aspects or problems), rules and constraints (e.g. no equal ranking), keywords as game pieces, flexibility (allowing rule-breaking), player-added elements, structured interaction
Process-oriented games	Co-design	Goals (negotiating and configuring care pathways), scenario and persona design, collaborative facilitation, sequenced question cards, digital or hybrid props (physical boards or Miro), roles (players as policymakers and designers), focus on iterative problem-solving

While structure is important in negotiations, it is equally essential to balance openness and constraint. The three types of negotiation games illustrate a purposeful use of game elements that help structure interactions while maintaining this balance. Game rules and objectives provide a framework for engagement, such as mapping actors, ranking priorities, or selecting problems, but flexibility is strategically incorporated throughout.

For example, in prioritization games, facilitators have the discretion to allow rule-breaking (e.g. ranking certain aspects equally), which maintains the negotiated nature of the interaction. This balance ensures that the dialogue is directed towards specific outcomes—such as validating findings or identifying problems—while participants still have the agency to challenge or reinterpret the analysis upon which the game is based.

We present our analysis of the case study in two parts. First, we detail our staging moves and our efforts to design and facilitate multiple situated negotiation games for co-exploration, co-analysis, and co-design, which are

integral to the facilitated negotiations. Second, we present our empirical findings, highlighting the contextual or situated nature of game development.

Staging negotiations using multiple negotiation games to engage healthcare practitioners and leaders in designing a cohesive healthcare system in Denmark

Round 1: visits to CHCs in Jutland

We kicked off the project by visiting 9 CHCs in Jutland, starting with those providing regional (midwife services, outpatient clinics, X-rays, blood analysis laboratories, etc.) and municipal (rehabilitation, nutrition experts, district nursing, etc.) services.

Staging initial visits

Each visit was staged according to the SNS framework to foster explorative negotiation between us as researchers and *relevant actors* (Figure 2, element 4) operating in the CHCs. We invited managers and health professionals from regions and municipalities into negotiation spaces at their premises, thereby ensuring a familiar *material arrangement* (Figure 2, element 3) for participants. We hoped that playing the games with them on their 'home field' would make them feel comfortable, and position them as 'hosts' informing us as 'guests' about their practices. *Interpreting and framing* (Figure 2, element a) these meetings using an exploratory approach allowed for open-ended negotiations in which participants articulated their perspectives and concerns about how their CHCs were (not) fostering increased collaboration between sectors.

We *inscribed* (Figure 2, element b) this intention and framing of cross-sectoral collaboration into a *negotiation game* (Figure 2, element 1) that focused on the collaborative aspects of CHCs while allowing participants to bring other related concerns to the table. To structure the dialogue, capture insights, and motivate actors to engage with us, we developed an A4-sized game board that illustrated the outline of a building and smaller game pieces that represented different actors and objects that we imagined would configure it: professionals (nurses, doctors, physiotherapists, administrative personnel, etc.) as well as non-human actors such as canteens, legislation, IT equipment, etc. The players were asked to map the central functions in their CHC (the goal) and to situate the game pieces to reflect its operations (rules of the game). In line with the open and exploratory nature of our framing, we planned to use a semi-structured *facilitation* (Figure 2, element 2) style to investigate whether a configuration game would help participants discuss and negotiate improvements to CHCs.

Facilitating and documenting the negotiations

The dialogue flowed naturally as the three researchers eagerly posed questions, and the participants were equally enthusiastic about discussing the collaboration challenges they faced in their CHCs. Instead of the participants interacting directly with the game pieces, one researcher documented their responses on the pieces and arranged them on the game board. This approach provided transparency during the negotiations and allowed participants to visually witness the evolution of their CHC based on their contributions. Additionally, the game format enabled researchers to take pictures, capturing a quick visual overview of the similarities and differences among CHCs for future analysis.

The completed game boards and game pieces clearly illustrate the variations in building configurations, services offered, types of ownership structures, and the efforts made to promote collaboration across different CHCs (see Figure 4).

Evaluating the game

Although it fostered transparency and enabled us to visually capture insights during the initial negotiation, the game had its challenges. First, the players did not interact with the game pieces themselves, which may be attributed to the *material arrangement* of the space. Oftentimes, dialogues took place at offices with large meeting tables, making it difficult for many participants and facilitators to see the relatively small material object (game board) which had been designed for fewer participants. Second, the game rules and the role of the facilitator had not been clearly explained to all three researchers, resulting in all of them being eager to pose questions along different



Figure 4. First design game illustrating the functions and actors of 4 CHCs.

tangents. Nevertheless, the game helped structure the dialogue and enabled the researchers to capture valuable insights.

Empirical findings: the municipality should be in charge

In most CHCs, collaboration was limited to monthly meetings involving municipal, regional, and specialized medical representatives. These meetings often focused on minor issues like budget allocations for repairs rather than citizen-focused cohesion. A key finding was that meetings shifted towards strengthening inter-sector collaboration when the municipality had ownership and primary responsibility for maintenance.

Generally, many CHCs faced the same challenges: (a) a lack of collaboration between health professionals in the building, (b) a lack of clarity about the value created by the CHC and its role in the healthcare system spanning the central hospital, multiple municipalities in the area, and civil society actors such as volunteers and GPs. Based on these findings, the project focus was slightly *reframed* to include hospitals, GPs, and members of civil society and the relations (or lack thereof) between these central actors.

Round 2: prioritizing and configuring the Danish healthcare sector

Reframing the project as well as the design game(s)

In addition to the filled-in game pieces, our empirical material comprised photos of negotiation games, notes, and audio files. We began coding everything to interpret our empirical findings and our evaluation of the initial game, and inscribed these into a new type of game to be played with managers and healthcare professionals in CHCs in Zeeland. This time, fewer people participated in the game. In some cases, we only had one player at a time. In other cases, we had multiple players—sometimes even from different sectors.

Based on our new project framing, we inscribed this *(re)framing* into two new negotiation games to investigate relations between CHCs, hospitals, municipalities and GPs, and between healthcare professionals and patients. Both games were much more straightforward with clear rules, and the game pieces were larger. Again, the spaces were staged and framed quite loosely to allow for co-exploration as invited actors from the CHCs voiced their concerns regarding the role of CHCs in the Danish healthcare system.

This time, a single facilitator was designated to initiate the games, emphasizing the aim of capturing participants' personal views on the Danish healthcare system.

The configuration game: Investigating relations between CHCs, hospitals, GPs, etc

The first game was a configuration game consisting of pieces representing central actors in the Danish healthcare system, such as hospitals, CHCs, GPs,

specialists, etc. (material constraints of the game; Figure 5, left side). We wanted participants to arrange these game pieces (the rules) to represent their view of the nature of a cohesive Danish healthcare system (the goal), including a particular focus on which role they wished to see the CHCs play. The designated facilitator introduced the rules, stating that players should arrange the game pieces freely to illustrate their view of the different actors in the healthcare system and explain why they configured the system the way they did.

In contrast to the initial game, this time, the participants quickly began arranging the game pieces while narrating their view of the system. We saw that they configured the game pieces differently depending on their position in the system, resulting in as many different interpretations of and visions for the Danish healthcare system as the number of participants we invited to play. Several GPs playing the game were interested in the relationship between patients and doctors regardless of whether interactions occurred in a CHC or elsewhere. Managers and healthcare professionals from municipalities understood the CHC as a central hub for collaborative efforts with people across municipal administrations (emphasizing that current CHCs allowed for enhanced collaboration between actors from their sector rather than across sectors). Regional professionals viewed the CHC as a central way of organizing across diverse professions in the region. Interestingly, all agreed that more collaboration between sectors was essential and much needed.



Figure 5. Configuration game (left) and prioritization game (right).

The prioritization game

Thereafter, a prioritization game was brought to the table. The second game (Figure 5, right side) was a ranking game designed to uncover the most critical aspects of a successful healthcare sector, focusing on relations between healthcare professionals and patients. It consisted of several keywords from previous interviews and strategy reports from regions and municipalities (e.g. cohesion, trust, comfort, locality, and accessibility). Inviting the players to co-analyse these findings, the facilitator asked the players to describe how they understood the words on the game pieces and to rank them by importance and add new ones if necessary. The only rule was that only one game piece could be placed in each row.

However, this rule was broken numerous times as players deemed several aspects equally important. One player said, 'I cannot choose since these are completely interrelated', pointing towards the triangle she had formed with the pieces labelled trust, quality and comfort. When the facilitator noticed the players resisting multiple attempts to separate the pieces, they allowed some rule-breaking. Because the game's purpose focused on fostering negotiations and dialogue rather than enforcing strict rules, participants were allowed to negotiate with the facilitator to break this rule.

Empirical findings

When analysing the results of the prioritization game, we scored the keywords based on their rankings from top to bottom (see Figure 6). Cohesion, or 'sammenhæng', stood out as the top priority for GPs and municipal and regional actors.

Evaluating the two games

The reduced complexity of the two games made interacting with them more manageable for healthcare professionals and managers. The game pieces were much bigger and easier to understand. Typically, only 1–2 relevant

	Cohesion	Local	Accessibility	Trust	Security	Quality	Service	Knowledge	User involv.	Compliance
Municipal actor	12	9	9	7	7	7	1	11	7	5
Municipal actor	10	7	7	3	5	5	5	5	9	10
Municipal actor	5	1	4	7	8	6	4	4		
Municipal + Regional actor	5	1	2	8	7	5	3	4	10	7
Regional actor	3	5	4	6	6	10	8	2	1	9
Regional actor	10	9	5	6	1	5	9	5		7
Regional actor	7	7	7	8	8	7	7	8		
Regional actor	4	1	2	7	3	6	5			
General Practitioner	6	1	2	4	4	6	8	6		
General Practitioner	8	4	1	7	7	6	1	3	6	
General Practitioner	8	1	2	7	7	7	4	3		
General Practitioner	8	7	7	3	3	3	7	4		
Total score	**86**	**53**	**52**	**73**	**66**	**73**	**62**	**55**	**33**	**38**

Figure 6. Result of ranking game. The green cells indicate that this aspect was ranked highest, while the orange ones were ranked lowest by that specific actor. Players added user involvement and compliance while playing the game.

actors and 1–2 facilitators were present, allowing everybody to see and participate in the games by voicing their ideas and concerns in relation to the keywords and different roles in the healthcare system. The simplicity of the games and the quantifiable nature of the results enabled us to compare negotiations and see that actors from all sectors of the healthcare system identified cohesion as a central parameter and that healthcare professionals across sectors viewed the CHCs as places to foster collaboration.

Round 3: five regional workshops, collaborative analysis of preliminary findings and co-creation of new relations between sectors

Until now, we had staged negotiations with actors from individual CHCs across the country. The project's next phase focused on staging and facilitating spaces for negotiations between managers and healthcare professionals from *multiple* CHCs within the same region in each of the 5 regions in Denmark. This was our chance to move from the exploratory phase of our project towards an analytic and generative phase. This time, we framed the negotiation space and, thus, our games much more narrowly. This also meant assigning a specific facilitator to each group playing the game.

Typically, we started the workshops by presenting our initial findings as a typology of CHCs in Denmark based on our empirical data. This provided participants with an intial opportunity to ask questions and either confirm or reject our preliminary findings. Afterwards, we facilitated two design games in smaller groups (typically 3–5 players, preferably from different sectors). The facilitator's primary role was to pose clarifying questions to encourage dialogue and ensure every participant had a chance to speak and voice their inputs and concerns. Secondarily, the faciliator monitored the time so that all cards could be addressed within the timeframe of the workshop.

Game one: current problems in the Danish healthcare system

For the first part of the workshop, we wanted to test our current understanding and investigate specific themes by staging a co-analysis of our initial findings. We interpreted our data from the first and second rounds of negotiations with actors from CHCs all over Denmark by coding transcriptions of all our previous interviews. During our coding, different themes emerged, such as central functions (x-ray, blood analysis, rehabilitation, etc.), ownership structures, and user types based on age, health conditions, digital competencies, health literacy, etc. We inscribed these insights into a game board and game pieces. The aim of the first game was to (a) facilitate a co-analysis of whether the CHC actors would see themselves reflected in our findings or challenge them, and (b) encourage the players to negotiate and agree on central problems collaboratively (and across sectors) (see Figure 7, left side).

We created a variety of cards (structuring game elements), each presenting a question in a specific order, and arranged them along the sides of the game board. For example: What is the most pressing issue in our current healthcare system? Who is impacted by this issue? Where in the system does it take place?

The first step of the game, selecting a problem to focus on (a rule), immediately spurred discussion. We presented five problems of which inequality, high complexity and lack of cohesion were selected most often across the workshops. As observed in the previous ranking game, the problems and topics were interlinked. For instance, a lack of compliance and a high number of readmissions could be linked to overarching challenges such as a lack of cohesion and high complexity. In some cases, each player wanted to focus on a different topic, typically resulting in a lot of negotiation before the group could select a problem. We interpret this as indicating that the players recognized each problem as important. Moreover, no new problems were identified during this stage, which led us to conclude that we had captured the most essential overarching problems in the Danish healthcare sector. During the rest of the game, the players unfolded the problem by collaboratively negotiating to answer the questions on each card and writing/drawing the answers on the game board (a structuring game element).

Game two: an ideal CHC

Building on the identification and elaboration of current issues in the healthcare system, the second design game adopted a more creative problem-solving approach. Participants shifted their focus to addressing the identified problems by designing and configuring an 'ideal' CHC. The game board maintained the same format as the first game for simplicity (see Figure 7, right side). This time, questions posed by the game included: What is the primary purpose of this new CHC? Which professionals and functions should be included? What types of interactions should occur between them?

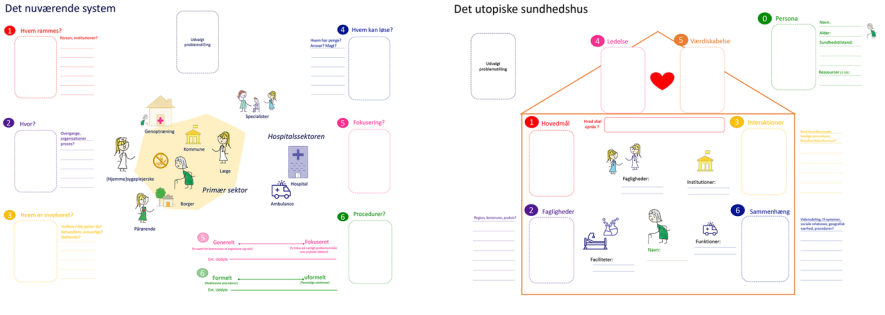


Figure 7. Game boards for the two design games focused on current problems (left), and how a CHC could be designed to accommodate and solve these problems (right).

Facilitation and empirical findings

Due to time constraints, very few groups managed to finish playing this game. Nevertheless, they all initiated discussions on which health services should be performed at the CHCs and which should be provided elsewhere, such as at the regional hospital. At this stage, the players took much more initiative without prompting from the facilitator, as they felt more comfortable with the game format. They agreed that the ideal building should include actors from various sectors and cross-sector teams. Furthermore, shared digital meetings and virtual cross-professional conferences were highlighted as flexible collaboration methods. Although engaging concepts such as 'the CHC that creates health and not merely the absence of disease' and 'the CHC that the patient/citizen requires' emerged, the most valuable aspect of playing this game, according to the players, was the conversations it ignited among participants from different sectors and professions. One player even requested an electronic copy of the game, expressing a desire to print it out and facilitate its use in her community.

Round 4: national online workshop

To conclude the research project, we staged a national conference, gathering participants from all over the country to develop new CHC concepts. We invited all previous informants and participants to visit our university campus, where we had booked our largest conference room for the event. We also encouraged our contacts to share the invitation with their networks to attract more attendees.

The objective of this workshop was twofold: (a) to give back to our collaborators by presenting our preliminary findings and facilitating co-analysis, and (b) to enable participants to negotiate while co-designing new solutions for the configuration of Danish healthcare, with a particular emphasis on the role of CHCs. We therefore arranged a programme consisting of presentations and a configuration game. In addition to ourselves, speakers included a representative from the Ministry of Health and a highly esteemed health economist.

Interpretation, framing and inscription of the game

In developing the final design game, we reviewed all the data and insights from the initial visits and regional workshops. Once again, we revisited the functions, ownership structures, and user types, which were validated as a result of the first game in the previous workshop. Our interpretation of this data led us to frame the workshop as a discussion on how to configure not only the CHCs, but the entire healthcare system. We aimed for the players to envision a healthcare system in which future CHCs could promote cohesion across actors and sectors.

Scenario- and persona-building based on empirical findings

Our data revealed significant regional differences across Denmark. To account for these differences and avoid one-size-fits-all solutions, we created a negotiation game with four scenarios to illustrate the various contexts and challenges in different parts of the country. For instance, one scenario focused on a rural area of Denmark, where many people are unemployed and there is a shortage of skilled healthcare professionals, particularly GPs. Another scenario focused on a larger city with more tech-savvy citizens and specialized CHCs. Two additional scenarios were developed, resulting in a total of four distinct scenarios.

To develop the game pieces, we compiled our insights into a matrix that resembled a morphological diagram, featuring an overarching topic and associated examples of solutions. For instance, we had a category called functions, with associated aspects such as x-rays, rehabilitation, and so on. We also had a category called frequency, with associated terms such as daily, monthly, on request, and so forth. However, some of the aspects were non-negotiable. In all four scenarios, the municipality owned the CHC and consequently held the main responsibility, as our data clearly demonstrated that municipal ownership was integral to successful collaboration within the building.

Days before the conference, Denmark was locked down due to COVID-19. The pandemic disrupted the setup and forced us to re-stage the physical workshop space by converting it to an online format. Instead of a material arrangement featuring tables, chairs, and a tangible design game, the workshop was streamed via Microsoft Teams, and the design game was facilitated using the online shared workspace platform Miro. This altered the entire configuration of the negotiation space, as we had to re-stage the space and convert the game board and all game pieces into digital representations (see Figure 8).

While the game board and pieces were relatively easy to transfer to the Miro board, assigning players to groups digitally and facilitating interactions among players who could not see each other's faces while playing proved significantly more challenging. We divided the participants into three groups, ensuring that group members represented different organizations and parts of the country, and designated ourselves as facilitators of the negotiation games. Each facilitator was assigned a specific scenario where participants aimed to re-design or re-configure the Danish healthcare system to suit local conditions. The focus was on fostering cohesion among the central actors and assessing where patients' interactions with the healthcare system should occur (in their homes, at the hospital, in municipal facilities, or at a CHC).

Playing and facilitating the game

Like in a physical game, we wanted the participants to move the game pieces and collaboratively configure the scenario with healthcare

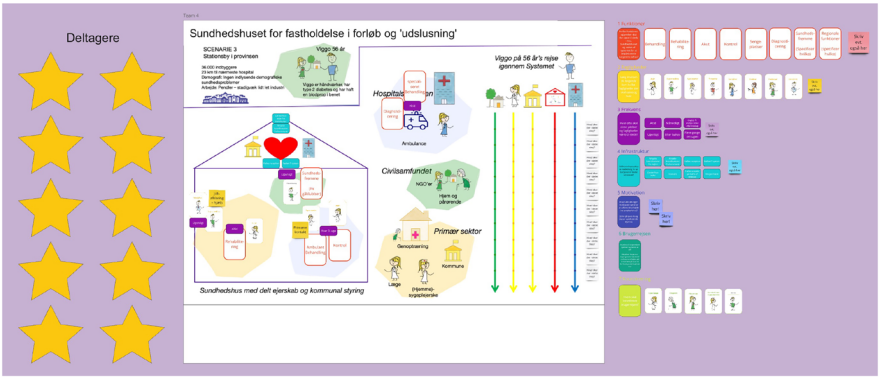


Figure 8. Online design game in Miro for the final workshop with space to write participants' names on the left, the game board in the Centre, and the game pieces on the right.

professionals, IT solutions, functions, frequencies for the availability of services, etc. However, some participants found interacting with the Miro platform to be challenging, as they were unfamiliar with this type of shared online workspace. Noticing that the players struggled to navigate the online platform, one of the facilitators quickly took charge and moved the game pieces according to the players' suggestions. A central task in any facilitation is to pose many questions to encourage participants to collaboratively reflect upon and negotiate every aspect of the game. Starting with the healthcare functions, the players were to reflect on whether rehabilitation should be situated at the CHC or a different location and how often the gym and physiotherapists should be available for the citizens. After each function was discussed and the cards placed, the facilitator simultaneously noted central quotes arguing for the choices made. Next up for negotiation was which professions were needed, where, and how often. The platform's transparency allowed the participants to verbally correct the facilitator if she had misunderstood something they said and helped her place the pieces on the game board where everyone agreed they should be positioned.

Empirical findings

As a result of the negotiations, three CHC concepts emerged. One was 'the CHC for less complexity', where the central idea is that not all functions are permanently housed in the CHC, but only on certain days or time slots. For instance, regional services such as outpatient treatment and regular examinations/check-ups need only be available on a bi-weekly or monthly basis, whereas GPs and municipal services, such as nurses and physiotherapists, should be available daily. Civil society organizations, such as patients' organizations, should also have access to rooms, e.g. after hours. A dedicated care pathway coordinator (possibly co-financed by the municipality and region) is

necessary to ensure coherence for citizens. It was suggested that a shared IT system accessible to every healthcare professional inside the CHC would be pivotal to ensuring coherence and coordination.

Because the game's online format created challenges that impeded players' direct interaction with the game board, the facilitator was given more responsibility for engaging participants in the negotiations, moving game pieces, and documenting players' ideas.

Epilogue: Mapping existing and co-designing new integrated care pathways

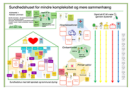
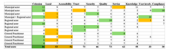
All the insights and reframings developed during the project were ultimately translated into more conventional material objects such as PowerPoint presentations, typologies, reports, and research articles. These were employed in negotiations we staged with public servants and other relevant actors to bring forward our findings on how to configure the healthcare system to support cohesive care pathways and influence current discussions regarding the development of the Danish healthcare system. The findings were also translated and inscribed into a more generic process-oriented negotiation game, formatted as a board game played with participants at a significant annual conference for healthcare leaders, healthcare professionals, IT companies, and policymakers. Once again, the aim of this game was twofold: (a) to foster negotiation among actors from different sectors, professions, and geographical locations, and (b) to co-design cohesive care pathways.

Overview of negotiation games played

As the focus of the research project shifted, so did the focus of the negotiation games. Initially, the emphasis was on understanding the existing setup and identifying the role of the current CHCs, along with the challenges preventing them from reaching their full potential. In contrast, the later games centred on co-designing future CHCs and their position within the healthcare system. This underscores the situated, iterative nature of negotiation games. Additionally, several types of games were used (see Table 1). Generally, most games concentrated on configuration, from configuring existing CHCs to envisioning the (re)configuration of the entire healthcare system. Other games focused on ranking, allowing for qualitative and quantitative data types. The final game was more process-oriented, shifting attention towards configuring aspects and creating cohesive care pathways across sectors.

Lastly, Table 2 summarizes how the focus of the negotiations evolves according to different moments in the process. From the outset, the emphasis was on co-exploration rather than co-analysis of the findings from previous negotiations, ultimately leading to co-design efforts. Thus,

Table 2. Overview of games: purpose, type, and focus.

	Round 1: Visits to CHCs in Jutland	Round 2: Visits to CHCs and on-on-one dialogue with GPs in Zeeland	Round 3: Five regional workshops (one in each region)	Round 4: National workshop	Epilogue: Mapping existing and co-designing new integrated care pathways
No. of visits	9 visits	12 visits	5 workshops	1 workshop	6 workshops
Relevant actors	Local municipal managers and healthcare professionals in Jutland (sometimes a single manager and sometimes groups of 3-6 people): CHC managers and employees such as nurses and physiotherapists	Local municipal managers and CHC employees such as physiotherapists, regional managers, and GPs	Regional and municipal managers and healthcare professionals located in the same geographical region of Denmark	CHC managers and healthcare professionals from all over Denmark as well as researchers, GPs, representative from Ministry of Health	1 IT in health conference with healthcare leaders, healthcare professionals, IT companies and policymakers 1 workshop with the Danish Agency for Digital Government 4 workshops in health communities in the Northern Region of Denmark
Material arrangement	Physical visits to the CHCs	Sometimes at the CHCs and sometimes in cafes when meeting with individual GPs not attached to a particular CHCs	3 in conference rooms at CHCs, 1 online, and 1 at the university	Online	Larger conference venues at hotels, hospitals, museums and a Danish agency, with room for up to 60 participants
Games	Configuration games: mapping current CHCs' functions 	Configuration game and prioritization game 	Configuration games: current problems in the Danish healthcare system and the ideal CHC 	Configuration game 	Configuration and process-oriented game 
Purpose why	Co-exploration	Co-exploration and co-analysis	Co-analysis and co-design	Co-analysis and co-design	Co-analysis and co-design
Facilitator	No direct facilitator; the 3 researchers posed questions to the relevant actiors who replied	1 primary facilitator (one researcher) and typically one player.	1 researcher served as the facilitator for each group	1 researcher served as the facilitator for each group	1 facilitator using a PowerPoint presentation to facilitate at a distance and keeping time, plus 1 facilitator answering questions
Outputs	Visual representations of the CHCs and written notes; more a mapping exercise than a game	Qualitative data about the role of CHCs and quantitative data on the ranking of potential benefits offered by CHCs 	Visual representations of current challenges in the Danish healthcare system and concrete ideas for CHC design	Concrete ideas for archtypes of CHCs; PowerPoint presentations, typologies, reports and research articles	Visual mappings of the complexity in the healthcare system as well as many ideas for how to improve processes and manage integrated care pathways

it became evident that the purpose, type, and focus of negotiation games are interconnected with their position in the design process. This is reflected in the researchers' staging efforts, which continually responded to their navigation among the varying agendas and concerns of the involved actors.

Discussion

In this section, we discuss how a staging approach is instrumental in developing negotiation games and how the purpose, type and focus of the games may be seen as a range of staging and navigational moves.

SNS may be used to develop and facilitate negotiation games

Pedersen (2020) argued that designers and researchers can gain valuable insights by staging and facilitating situated negotiations. Our case study illustrates how carefully developed design games are key to encouraging reflection, facilitating negotiations, aligning agendas and concerns, and building consensus among healthcare professionals from different sectors, as well as between healthcare professionals and designers. The combination of SNS and negotiation games fosters double-loop learning (Schön 1984), allowing empirical findings from one space to inform and be inscribed into the next.

Typically, games (especially policy games) are designed to be played at stand-alone workshops. However, the situatedness of the SNS framework can be utilized by designers, researchers, and health professionals to understand development processes as comprising a series of negotiation spaces, with different types of tailored negotiation games developed and facilitated to fit the specific situation, following and building on one another (see Table 1). This assigns a prominent role to the negotiation games in making sense of empirical findings, as they represent the current project framing and the views and concerns of the participants. For instance, the initial focus on functions and technologies inside CHCs was later translated into a more systemic view of the interplay between CHCs, hospitals, municipalities, and GPs. In this sense, the gameboards represented the situated framing of the negotiations and the project scope. After the games were played, the gameboards, adorned with writing and drawings, transitioned to representing the views and concerns of the players, thereby contributing to the (re)framing of the ongoing development or research process. These findings enhance the SNS framework by emphasizing the central role of negotiation games, their types, and their capacities for co-exploration, co-analysis, and co-design.

Co-analysis, co-design, co-exploration: Incorporating co-analysis into participatory design

As the project progressed, the project scope and game design were continuously reframed based on previous negotiations. This meant that different types of games were used throughout the process for different purposes. Even though the main focus of every game was to enable, facilitate, and structure negotiations, they were designed to fit the overall project phases of exploration, analysis, and development. Since the research project was collaborative, the negotiation games focused on (a) co-exploration, (b) co-design, and (c) co-analysis.

Exploration and co-design are already associated with design games in the participatory design literature, (Brandt, Messeter, and Binder 2008). For example, Kjaersgaard et al. (2021) explicitly investigated explorative aspects since they primarily used design games in their fieldwork. Likewise, conceptualisation-related games focused on co-designing possible futures as a response to identified problems have been described in the design game literature (Brandt, Messeter, and Binder 2008; Pirinen and Tervo 2020). In addition to these aspects, negotiation games enable us to investigate *co-analysis*. Co-analysis is already used in some form in policy games; however, in this case, we see several collaborative analyses as a central framing used by researchers to continuously test their gathered empirical data, allowing for reinterpretation of previous insights. For instance, the researchers translated their findings of significant challenges in the healthcare sector, such as lack of cohesion and inequality, into game pieces that would become key elements in prioritization, configuration, and process-oriented games. In co-analysis, games bridge diverse perspectives and make analytical elements explicit, enabling informants to solidify, substantiate, or falsify the analytically generated insights on which the games are based. This highlights the situatedness of negotiation games. They lock certain parameters, such as predefined game pieces (e.g. central actors or keywords) or a specific scenario, to focus discussions while leaving the interpretation of relationships, priorities, and configurations open for participant-driven negotiation. Card drawing was introduced to steer the focus throughout the interaction and as an element of unpredictability to prevent participants' preconceived notions from taking over the game's focus. This interplay between structure and flexibility supports designers' and players' deep engagement with empirical findings and fosters shared understanding across sectors.

Thus, we argue that negotiation games draw upon inspiration and experience with both design games and policy games, simultaneously linking the development of tailored games to the staging and navigational efforts involved in design-led research in a healthcare setting.

Configuration, prioritization, and process-oriented games

Most of the games were configuration-based. They were developed from a desire to learn about the current situation at CHCs and the healthcare system and later to situate our ongoing analytic insights within the participants' daily context. Configuration games facilitate the mapping and reflection of current practices and roles, thus serving as a baseline for discussing the (re)configuration of future scenarios. Prioritization games yielded both qualitative data (e.g. interpreting the meanings related to the ranked topics) and quantitative data (i.e. assigning scores to the themes based on the rankings). The process-oriented aspect emerged in the later stages of the project as researchers shifted their focus towards patient experiences (i.e. cohesive care pathways across sectors), offering a designerly perspective on more traditional patient journey mapping tools, such as those described by McCarthy et al. (2016)

Are the games really games?

Although they cannot be described as traditional games, the negotiation games in this project intentionally incorporated game-like elements to structure interactions and balance openness with constraints. Early configuration games employed boards and pieces to map CHCs, but depended heavily on facilitators, which limited participant engagement due to the complexity and size of the game pieces. Subsequent iterations addressed this limitation by introducing larger and more tangible game pieces, and encouraging players to manipulate them physically, draw, and write directly on the boards and pieces. This adjustment enabled participants to actively co-construct insights rather than passively observe the process. Prioritization games introduced structured constraints, such as ranking keywords, while allowing flexibility for rule-breaking that enabled participants to challenge analytical framings. In the final processual games, predefined scenarios focused discussions; however, the game posed challenges to players' participation due to their lack of experience with the online format, which hindered their interaction. The facilitator intervened and mediated the negotiations by manoeuvring the game pieces based on the players' guidance.

Although elements of design games such as turn-taking or chance (Brandt 2006) were absent, facilitators introduced unpredictability and ensured inclusivity through dynamic questioning. The negotiation games served as boundary objects, stipulating certain parameters (e.g. predefined actors or scenarios) while leaving relationships and priorities open for negotiation. Thus, they effectively supported co-analysis and bridged diverse perspectives. An element like turn-taking allows each player to voice their

opinions, but it was not integrated into the negotiation games because a facilitator was present to ensure every participant had a chance to speak.

Conclusion

This research has shown that negotiation games can foster co-exploration, co-analysis, and co-design among researchers, designers, and healthcare professionals who sometimes struggle to find common language and motivations. In particular, their capacity to encourage co-analysis by bridging diverse perspectives and making analytical components explicit enables players to solidify, substantiate, or falsify the analytically generated insights upon which the games are based.

Moreover, an SNS approach is instrumental in developing and facilitating these games and processes. Our findings illustrate that negotiation games developed using the SNS framework encourage reflection, facilitate negotiations, align agendas, and build consensus among healthcare professionals across sectors as well as between healthcare professionals and designers. This iterative approach enables double-loop learning, enabling empirical findings from one space to inform and be inscribed into the next. This process was reflected in the progression from initial configuration games focused on mapping current practices to prioritization and process-oriented games that supported co-analysis and co-design.

This research contributes to the SNS framework by emphasizing the central role of negotiation games in co-creating solutions and navigating complex healthcare challenges. The games' capacity to represent the situated framing of negotiations and the project scope highlights their value as tools for fostering alignment and collaborative problem-solving in healthcare design processes.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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